Lab Experiment 4: JavaScript Arrays, Functions, and AI Assistance

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Learning Outcomes

By completing this lab experiment, students will be able to:

- Create and manipulate JavaScript arrays
- Use loops to process array data
- Create reusable JavaScript functions
- Use conditional statements for decision-making
- Apply AI tools to assist with debugging and code generation
- Display formatted results in the console

Activity Description

Write a complete JavaScript program that demonstrates the use of arrays, loops, functions, and conditional statements.

Students are encouraged to use AI tools to:

- Generate code ideas
- Debug syntax errors
- Improve code formatting
- Understand JavaScript logic

Save the .js file using your last name.

Example:

`smith.js`

Include an information header at the top of the file.

Information Header Example

```
/*
Name: John Smith
Course: JavaScript Programming
Lab Experiment 4: Arrays, Functions, and AI Assistance
Date:
*/
```

Class Example 1: Creating an Array

Example Code

```
console.log("\n------ Class Example 1: Arrays -----");

let fruits = ["Apple", "Banana", "Orange"];

console.log(fruits);
console.log("First fruit:", fruits[0]);
```

Class Example 2: Using a Loop with Arrays

Example Code

```
console.log("\n------ Class Example 2: Loop Through Array -----
");

let colors = ["Red", "Blue", "Green"];

for (let i = 0; i < colors.length; i++) {
    console.log(colors[i]);
}
```

Class Example 3: Functions and Return Values

Example Code

```
console.log("\n------ Class Example 3: Functions -----");

function squareNumber(num) {
    return num * num;
}

console.log("Square:", squareNumber(5));
```

Lab Exercise Student Score Analyzer with AI

Instructions

Create a JavaScript program that:

1. Declares an empty array named scores
2. Uses a for loop to collect five student scores using prompt()
3. Stores the scores in the array
4. Creates a function named calculateAverage() that returns the average score
5. Uses conditional statements to determine:

    - Passing average
    - Failing average
6. Displays the following in the console:
    - All entered scores
    - The average score
    - Pass or fail message

Requirements

- **Passing Rules**
    - Average score: 70 or higher, output: class passed
    - Average score: below 70, output: class failed

Sample Output

```
Scores: 80,90,75,60,85
Average Score: 78
Class Passed
```

AI Integration Activity

Instructions

Use an AI tool to assist with one of the following:

- Debugging your code
- Improving variable names
- Explaining a loop or function
- Generating additional test cases

After using AI, add a comment in your .js file describing how AI helped you.

Example

```
// AI Assistance:
// ChatGPT helped explain how the calculateAverage()
// function works and assisted with debugging.
```

Submission Requirements

- Complete the class examples and lab exercise in one .js file
- Upload the completed .js file to your GitHub repository
- Submit the GitHub repository HTTPS link

Grading Criteria

- Class examples: 3 pts
- Lab Exercise: 4 pts
- AI integration Activity: 3 pts